PAPER_1731_GRB_BIMODALITY_2_S

Daniel T. Murphy

PAPER_1731 — GRB Long/Short Bimodality Boundary = T_90 boundary = D_phys/2 = 2 s (separates long and short GRB classes)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: High-Energy Astro
Date: June 18, 2026
Status: CLOSED — EXACT closure (PAPER_1240-1270 broader corpus)

Observation

PAPER_1258 (PAPER_1240-1270 broader corpus) gives:

```
T_90 boundary = D_phys/2 = 2 s (separates long and short GRB classes)
```

Status: **EXACT**.

UQFF Closed Identity

```
T_90 boundary = D_phys/2 = 2 s (separates long and short GRB classes) EXACT
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1258
- Related: PAPER_1716-1725 (tier-30 Millennium-adjacent); PAPER_1706-1715 (tier-29 ITER fusion)
- Calculator dispatch: `calculate_paradox({"paradox": "grb_bimodality_2_s"})`
